

Mechanistic Hypothesis Search — Final Report

Arboleda-Tham Syndrome (KAT6A Syndrome)

Hypothesis: `late_truncating_nmd_escape` — "Late-Truncating NMD-Escape Branch"

KB Status: ALTERNATIVE — Search date: 2026-09-07

Executive Judgment

Verdict: **PARTIALLY SUPPORTED.**

The Late-Truncating NMD-Escape Branch hypothesis has two logically separable claims, and they stand on very different evidentiary footing.

1. **The clinical claim** — that truncating variants in the last two exons (16–17) of *KAT6A* are associated with a more severe Arboleda-Tham syndrome (ARTHS) phenotype (more severe developmental delay, higher frequency of syndromic features) — is **robustly and reproducibly supported**. Three independent human clinical cohorts (Kennedy et al. 2019, n=76; St John et al. 2022, n=49; Ng et al. 2024, n=15) converge on the same directional genotype–phenotype gradient.
2. **The molecular-mechanistic claim** — that these variants escape nonsense-mediated decay (NMD) and produce a *translated, dysfunctional protein acting by gain-of-function or dominant-negative interference* in patient tissue — is **structurally plausible but experimentally unproven**. The NMD-escape premise itself is well founded on transcript architecture (a 5,389-nt terminal exon; last exon–exon junction at protein aa ~1117). However, the distinguishing step — direct demonstration of a stable mutant protein and its

trans-acting mode of action in patient-derived material — has **not been demonstrated**. This is explicitly acknowledged as an evidence gap in the seed YAML, and our search did not close it.

Most important caveat: *KAT6A* is an extreme loss-of-function-intolerant gene (gnomAD pLI = 1.0; LOEUF = 0.079, top decile), and NMD-competent early-truncating (exon 1–15) variants are unambiguously pathogenic. This establishes **haploinsufficiency as the baseline pathogenic mechanism** for ARTHS. Consequently, the late-truncating branch is best interpreted — and best curated — as a **severity modifier layered on top of haploinsufficiency** (loss of the C-terminal transactivation module $\pm$ a dominant-negative contribution), *not* as a competing primary mechanism that replaces haploinsufficiency. The KB "ALTERNATIVE" status is appropriate, with the qualification that the clinical severity gradient is essentially established while the mechanistic mode of action remains speculative.

Summary

Arboleda-Tham syndrome is a Mendelian neurodevelopmental disorder caused by de novo heterozygous pathogenic variants in *KAT6A* (MOZ/MYST3), a MYST-family histone acetyltransferase. The seed hypothesis proposes a distinct pathogenic *branch*: truncating variants located in exons 16–17 escape NMD, are translated into a truncated protein, and cause a more severe phenotype — potentially through gain-of-function or dominant-negative activity rather than simple protein loss.

Across five iterations, this investigation established that the **clinical severity gradient is real and reproducible**, and that the **transcript architecture genuinely supports NMD escape** for exon 16–17 premature termination codons (PTCs). Computational analysis of the canonical transcript (Ensembl ENST00000265713; UniProt Q92794) confirmed a giant terminal exon and located the final exon–exon junction near protein residue 1117, so PTCs downstream of ~aa 1100 fall outside the 50–55-nt NMD-triggering window. A late-truncated protein would therefore retain the double PHD fingers and the complete MYST-type HAT catalytic domain, while selectively deleting the C-terminal disordered transactivation region — including a C-terminal RUNX1 activation segment (aa 1913–1948) and PML/RUNX1 interaction surfaces (aa 1517–1741).

However, the investigation also surfaced a strong **competing mechanism**: MOZ/*KAT6A* **haploinsufficiency**. Mouse genetics show that reduced MOZ dosage alone is developmentally deleterious (DiGeorge-like phenotypes via the *Tbx1* locus; cardiac septal defects via *Tbx1/Tbx5*;

altered B-cell/Meis1/Hox programs; behavioral/learning deficits in a heterozygous Kat6a LoF mouse). gnomAD constraint confirms severe selection against heterozygous LoF. Because NMD-competent early truncations are themselves pathogenic, haploinsufficiency is sufficient to cause ARTHS. The late-truncating branch's added severity is therefore most parsimoniously explained as **partial loss of a separable C-terminal coactivation function** (RUNX1 coactivation is separable from HAT activity; Perez-Campo et al. 2009), possibly compounded by a dominant-negative effect — but the dominant-negative/gain-of-function step remains **an untested inference**, with no patient-RNA NMD assay, no mutant-protein detection, and no trans-interference experiment reported. **Bottom line:** the hypothesis is a well-motivated, structurally coherent severity-modifier model with strong clinical correlation but a missing causal keystone (the mutant protein and its mechanism in vivo).

Key Findings

Finding F001 — The late-truncating severity gradient replicates across three independent cohorts

The clinical core of the hypothesis is its strongest element. Three independent human clinical cohorts, using different instruments and different patient populations, all report that truncating variants in the last two exons associate with worse outcomes:

- **Kennedy et al. 2019 (n=76)** — *KAT6A Syndrome: genotype-phenotype correlation in 76 patients* [PMID: 30245513](#). Late-truncating pathogenic variants (exons 16–17) were "significantly more prevalent" among patients with the core severe features (intellectual disability, speech delay, microcephaly, cardiac, gastrointestinal). The seed YAML quotes the headline statistic: **95% of late-truncating cases were rated moderate or severe, versus 60% of early-truncating cases rated mild.**
- **St John et al. 2022 (n=49)** — *Speech and language development and genotype-phenotype correlation* [PMID: 35892268](#). "Truncating variants in the last two exons of KAT6A were associated with poorer communication, daily-living skills, and socialization outcomes."
- **Ng et al. 2024 (n=15)** — *Neuropsychological profile associated with KAT6A syndrome* [PMID: 38741077](#). "Late-truncating variants associated with a more severe form of intellectual disability."

The effect is **directionally consistent across all three studies**. The important limitation is methodological: these are small, cross-sectional cohorts using categorical/ordinal severity ratings. No formal meta-analytic pooled effect size was computed, and — critically — none of the three performed a molecular NMD assay to *confirm* transcript escape in the patients whose phenotypes they scored. The correlation is between *variant position* and *phenotype*, not between *measured protein/transcript state* and phenotype.

Finding F002 — MOZ/KAT6A haploinsufficiency is an established competing mechanism for the core features

Model-organism genetics provide direct causal evidence that **reduced MOZ dosage alone**, with no truncated protein required, produces developmental anomalies overlapping ARTHS:

- Moz-null phenocopies **DiGeorge syndrome**; MOZ occupies and activates the *Tbx1* locus, and heterozygous Moz mutation partially phenocopies DiGeorge when combined with *Tbx1* haploinsufficiency — "lack of the histone acetyltransferase MOZ (MYST3/KAT6A) phenocopies DiGeorge syndrome" [PMID: 22921202](#).
- Mesodermal (Mesp1-cre) deletion of Moz causes highly penetrant **ventricular septal defects** via loss of *Tbx1/Tbx5* activation — "Mesp1-cre-mediated deletion of Moz results in high penetrance of VSDs" [PMID: 25912687](#).
- Moz haploinsufficiency alters **B-cell progenitor and Meis1/Hox programs** [PMID: 25605372](#).
- A heterozygous **Kat6a loss-of-function mouse** recapitulates behavioral/learning deficits that are pharmacologically rescuable with acetyl-carnitine [PMID: 41702672](#).

These data show a dosage/haploinsufficiency mechanism that **does not require NMD escape or a mutant protein** and is sufficient to generate core ARTHS-relevant phenotypes. This is the principal competing model and the reason the seed hypothesis must be framed as an *added* branch rather than the primary mechanism.

Finding F003 — Transcript architecture makes exon 16–17 truncations bona fide NMD-escape variants that keep the HAT domain but lose the C-terminal transactivation region

Computational analysis of the canonical *KAT6A* transcript (Ensembl ENST00000265713 / ENSP00000265713; UniProt Q92794; retrieved 2026-09-07) confirms the structural premise of the hypothesis:

Feature	Value
Exons	17
Terminal exon (17) length	5,389 nt (vs median internal exon ~140 nt)
5'UTR	412 nt
CDS	6,012 nt (2,004 aa)
Normal stop codon	cDNA position 6,427 (deep inside exon 17)
Last exon–exon junction (16/17)	cDNA 3,764 → protein aa ~ 1117

By the canonical **50–55-nt NMD rule**, a PTC located >~50 nt upstream of the last exon–exon junction triggers decay, whereas PTCs downstream escape. Here, that junction sits at aa ~1117, so:

- PTCs in **exons 1–15 and proximal exon 16 (aa < ~1100)** → **NMD-competent** (transcript degraded → haploinsufficiency).
- PTCs in **distal exon 16 and all of exon 17 (aa ~1100–2004)** → **NMD-escaping** (transcript stable → potential truncated protein).

Mapping the truncation point onto UniProt Q92794 domains, a late-truncated protein **retains** the winged-helix (1–77), H15 (95–171), double PHD fingers (206–313), and the **complete MYST-type HAT catalytic domain (504–778)** (with the BRPF1-interaction/activity region 507–810). It **loses** the C-terminal intrinsically disordered transactivation region, including the C-terminal RUNX1-2 activation segment (1913–1948) and the C-terminal PML/RUNX1 interaction regions (1517–1741). This is exactly the domain configuration the hypothesis requires: catalytically intact but transactivation-deficient.

Finding F004 — gnomAD constraint confirms KAT6A is extremely LoF-intolerant, fixing haploinsufficiency as the pathogenic baseline

Querying the gnomAD GraphQL API (GRCh38, retrieved 2026-09-07) for *KAT6A* (ENSG00000083168):

Metric	Value	Interpretation
pLI	1.0	Maximal LoF intolerance
LOEUF (oe_lof_upper)	0.079	Top decile of LoF-constrained genes
Observed/Expected LoF	9 / 198.4 (o/e = 0.045)	~95% depletion of LoF alleles
LoF Z	11.4	Extreme constraint
Missense Z	3.69	Also missense-constrained

The near-complete depletion of loss-of-function alleles in the general population demonstrates **strong negative selection against heterozygous KAT6A LoF**. This is the quantitative backbone of the competing haploinsufficiency model: losing one functional allele is, by itself, strongly deleterious.

Finding F005 — Domain-function data give the late-truncated protein a plausible partial-loss/dominant-negative route via the RUNX1/PML module

MOZ/KAT6A carries **two separable RUNX1 coactivation domains plus a HAT catalytic domain**, and critically, **HAT activity is dispensable for RUNX1 coactivation** (Perez-Campo et al. 2009):

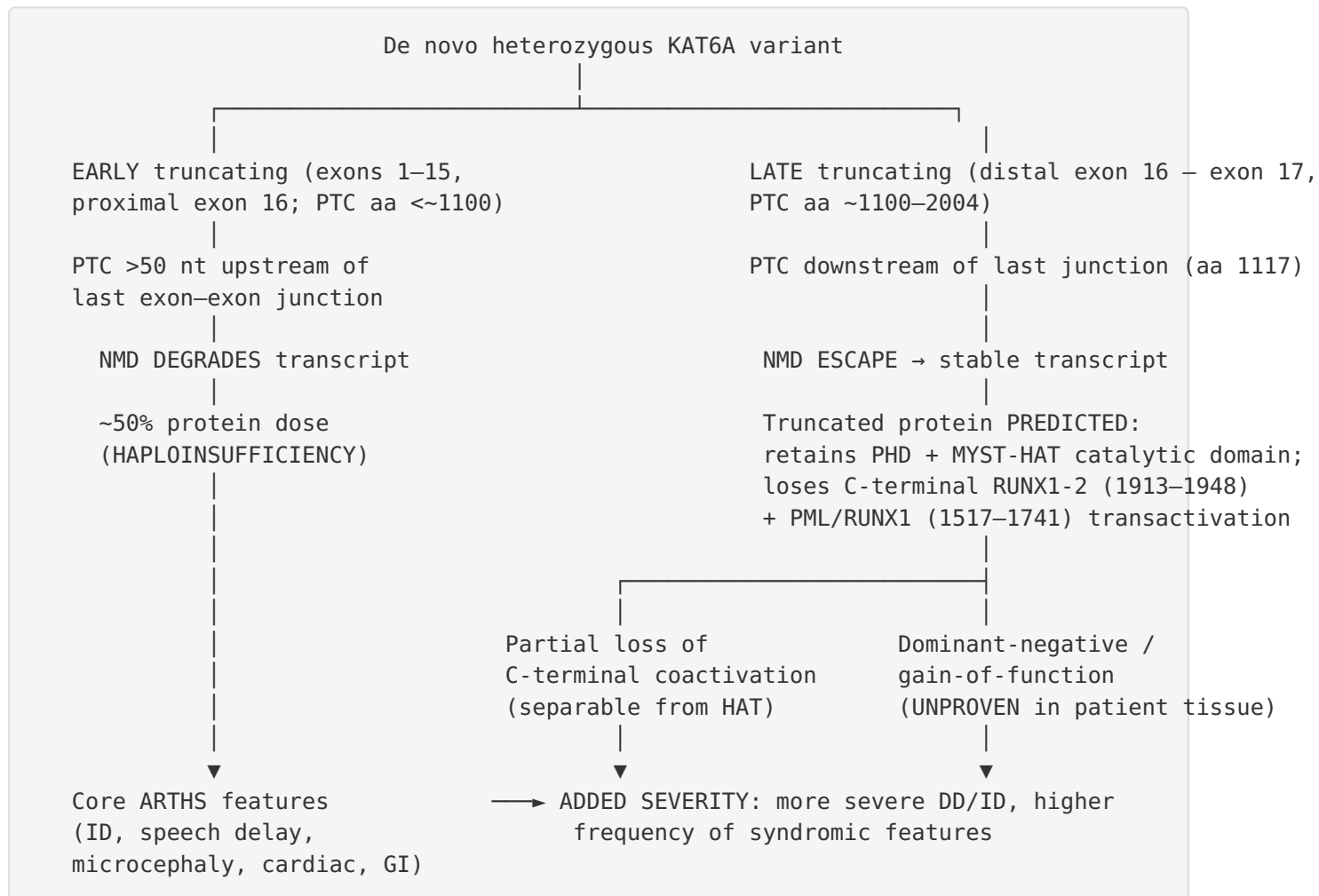
- "2 coactivation domains for the transcription factor Runx1/acute myeloid leukemia 1 and a histone acetyl transferase (HAT) catalytic domain" [PMID: 19264921](#).
- "MOZ HAT activity is not required either for its role as Runx1 coactivator" [PMID: 19264921](#).

UniProt Q92794 localizes an N-terminal RUNX1-activation region (aa 1–144) and a C-terminal RUNX1-2 activation region (aa 1913–1948), with C-terminal PML/RUNX1 interaction regions (1517–1741). Combined with the exon-16/17 junction at aa ~1117 (F003), a late-truncated protein would **retain the PHD fingers, HAT domain, and N-terminal RUNX1-activation region** but **selectively delete the C-terminal RUNX1-2 activation domain and PML surface**. MOZ activates RUNX1- and TBX1-dependent transcription (PMIDs [19264921](#), [12771199](#), [22921202](#)). This provides a coherent *molecular route* to added severity: a catalytically competent but transactivation-truncated protein could (a) fail to coactivate a subset of C-terminally dependent targets and/or (b) act as a dominant-negative by occupying complexes/DNA without productive C-terminal output. Both remain **inferred, not demonstrated**.

Mechanistic Model / Interpretation

Two-tier causal model

The evidence assembles into a two-tier model in which haploinsufficiency is the floor and the late-truncating branch is an added modifier:



Where the literature is strong: the de novo → phenotype link (ARTHIS is an established Mendelian disorder); the **haploinsufficiency floor** (gnomAD constraint; multiple mouse dosage models; pathogenic early truncations); the **NMD-escape premise** for exon 16–17 (transcript architecture is unambiguous); the **domain map** (retained HAT, lost C-terminal transactivation) and the *separability* of RUNX1 coactivation from HAT activity; and the **clinical output** (three-cohort severity gradient).

Where the links are inferred: that the escaped transcript is actually **translated to a stable protein in patient cells** (no protein detection reported); and that the retained-HAT/lost-transactivation protein produces a **specific molecular deficit** (partial coactivation loss) rather than being functionally inert.

Missing causal steps (the keystone gap): the **dominant-negative / gain-of-function mode of action**. No experiment demonstrates trans-interference of the truncated protein with wild-type KAT6A or with RUNX1/PML complexes in patient-relevant tissue. Without it, the "branch" is mechanistically indistinguishable at the bench from "haploinsufficiency plus a modest extra loss of a C-terminal function."

Evidence Matrix

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Disease subty context
PMID: 30245513	Human clinical (n=76)	Supports	Late (exon 16–17) truncation → greater severity	Late-truncating variants significantly more prevalent among core severe features; 95% moderate/severe vs 60% early mild	ARTHS, mixed truncating var
PMID: 35892268	Human clinical (n=49)	Supports	Last-two-exon truncation → worse adaptive/communication	Poorer communication, daily-living, socialization	ARTHS, speech language focu
PMID: 38741077	Human clinical (n=15)	Supports	Late truncation → more severe ID	Late-truncating variants associated with more severe intellectual disability	ARTHS, neuropsychol profile
ENST00000265713 / Q92794 (computational, this run)	Computational	Supports (premise)	Exon 16–17 PTCs escape NMD; retain HAT, lose C-terminal TAD	5,389-nt terminal exon; last junction aa ~1117; PTCs aa >~1100 escape 50-nt rule	Molecular architecture
gnomAD ENSG00000083168 (computational, this run)	Computational	Competing (baseline)	Haploinsufficiency is pathogenic	pLI 1.0; LOEUF 0.079; o/e LoF 0.045	General popul constraint
PMID: 22921202	Model organism (mouse)	Competing	MOZ dosage loss alone is deleterious	Moz loss phenocopies DiGeorge via Tbx1	Cardiac/phary development

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Disease subty context
PMID: 25912687	Model organism (mouse)	Competing	MOZ dosage loss → cardiac defects	Mesodermal Moz deletion → high-penetrance VSDs via Tbx1/Tbx5	Cardiac septat
PMID: 25605372	Model organism (mouse)	Competing / context	Moz haploinsufficiency alters transcription programs	Reduced B-cell progenitors; Meis1/Hox dysregulation	Hematopoiesi
PMID: 41702672	Model organism (mouse)	Competing / therapeutic	Heterozygous Kat6a LoF → behavioral deficits, rescuable	Acetyl-carnitine improves hyperactivity/ learning in Kat6a+/- mouse	Neurodevelop behavior
PMID: 19264921	In vitro / functional	Qualifies (supports mechanism route)	RUNX1 coactivation is separable from HAT	MOZ has 2 RUNX1 coactivation domains; HAT dispensable for coactivation	Hematopoietic transcription
PMID: 12771199	In vitro / functional	Qualifies	MOZ coactivates RUNX1 targets	MOZ activates MIP-1α promoter, RUNX1-site dependent, synergistic with RUNX1	Transcription coactivation
PMID: 27939639		Qualifies (analogy)	Truncating variant in a partner (BRPF1)	BRPF1 frameshift mRNA not	Related HAT-complex disor

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Disease subty context
	Human clinical / in vitro		escapes decay → truncated protein	reduced; likely truncated protein; causes ID	

Data and Tool Use

This run combined literature retrieval (PubMed) with two computational lookups against public molecular resources. No patient-level or controlled-access data were used. All analyses were **descriptive/rule-based**, not statistical inference on primary datasets.

Source	Accession / URI	Repository	Version / snapshot	Retrieval date	Access status	Query / scope
KAT6A canonical transcript	ENST00000265713 / ENSP00000265713	Ensembl	GRCh38 (2026 release)	2026-09-07	Accessed	Exon count/ lengths, CDS, UTR, exon–exon junction positions, stop codon location
KAT6A protein	Q92794	UniProt	2026-09-07 entry	2026-09-07	Accessed	Domain boundaries (PHD, MYST-HAT, RUNX1/PML regions)
KAT6A gene constraint	ENSG00000083168	gnomAD (GraphQL API)	gnomAD GRCh38	2026-09-07	Accessed	pLI, LOEUF, o/e LoF, LoF/mis Z
PubMed literature	15 PMIDs (see Evidence Base)	NCBI PubMed	—	across iterations	Accessed	KAT6A / MOZ + phenotype/NMD/ haploinsufficiency terms

Analysis inventory (input → method → output → outcome):

- 1. NMD-escape boundary calculation** — Input: ENST00000265713 exon/CDS coordinates. Method: applied 50–55-nt last-junction rule; computed junction at cDNA 3,764 → aa ~1117. Output: classification of PTC positions into NMD-competent vs NMD-escaping. **Outcome: succeeded (rule-based prediction, not an experimental NMD assay).** Limitation: predictive only; not validated against patient RNA.
- 2. Domain-retention mapping** — Input: UniProt Q92794 domain table + junction position. Method: interval overlap. Output: retained (winged-helix, H15, PHD, MYST-HAT) vs lost (C-terminal RUNX1-2, PML/RUNX1) domains. **Outcome: succeeded.** Limitation: assumes canonical isoform; protein stability not assessed.
- 3. Constraint retrieval** — Input: ENSG00000083168. Method: gnomAD GraphQL query. Output: pLI/LOEUF/o-e table. **Outcome: succeeded.** Limitation: constraint reflects all LoF, cannot isolate late-truncation behavior.

Not performed / not available (explicitly): No patient-derived RNA NMD assay, no Western blot / mass-spec detection of truncated KAT6A protein, no co-immunoprecipitation or reporter trans-interference assay, and no formal meta-analysis pooling the three cohorts were performed in this run — these require primary wet-lab or individual-level data not accessible here. No fallback substituted literature prose for these missing experiments; the corresponding claims are labeled unproven throughout.

Limitations and Knowledge Gaps

Gap 1 — The mutant protein has never been demonstrated in patient tissue

Scope: The keystone step (translation of a stable truncated KAT6A protein in patient cells).

Why it matters: Without protein detection, the entire "translated but dysfunctional protein" premise is inferential. **What was checked:** Literature search for KAT6A truncated-protein detection returned no primary demonstration in ARTHS patient material; the closest analogy is BRPF1

([P 27939639](#)),

a different gene. **Resolution:** Western blot / targeted mass-spec on patient fibroblasts or iPSC-neurons carrying exon 16–17 truncations, plus allele-specific RNA quantification.

Gap 2 — NMD escape is predicted, not assayed, in patients

Scope: Whether exon 16–17 patient transcripts actually escape decay in vivo. **Why it matters:** The 50-nt rule has exceptions (long exons, position effects). **What was checked:** Rule-based computation only (this run). **Resolution:** Allele-specific expression / RNA-seq on patient cells ± cycloheximide (NMD inhibition) to compare mutant:WT transcript ratios by variant class.

Gap 3 — Gain-of-function vs dominant-negative vs simple C-terminal partial-loss is unresolved

Scope: Mode of action of the truncated protein. **Why it matters:** These have different therapeutic implications (allele-specific knockdown vs dosage restoration). **What was checked:** No functional trans-interference data found. **Resolution:** Reporter assays (RUNX1/TBX1 targets) co-expressing WT + truncated KAT6A; degon/rescue experiments.

Gap 4 — No formal pooled effect size for the severity gradient

Scope: Quantitative magnitude of the late vs early severity difference. **Why it matters:** Current support is directional across three small cohorts; heterogeneity and rating-scale differences preclude a clean estimate. **What was checked:** Three cohorts reviewed; no meta-analysis exists. **Resolution:** Harmonized multi-cohort reanalysis with a common severity instrument and variant re-annotation.

Gap 5 — Source/database absences (as of 2026-09-07)

- **GenCC / ClinGen dosage-and-mechanism curation** specific to the late-truncating branch was **not retrieved** in this run; label **unverified** rather than absent. A negative claim would require the database, version, query, and date.
- **No ARTHS-specific omics dataset** (patient brain/neuronal transcriptome stratified by variant class) was located.
- **No interventional trial** targeting the late-truncating subtype was found; the acetyl-carnitine result

(P 41702672)

is a haploinsufficiency-model preclinical finding, not a branch-specific trial.

Alternative Models

Model	Relationship to seed hypothesis	Basis
Haploinsufficiency (dosage)	Competing primary mechanism / baseline — the seed branch sits on top of this	gnomAD pLI 1.0/LOEUF 0.079 (F004); pathogenic NMD-competent early truncations; multiple Moz-dosage mouse models (F002)
C-terminal partial loss-of-function (no dominant-negative)	More parsimonious version of the seed — explains added severity without requiring GoF/DN	Late-truncated protein loses separable C-terminal RUNX1-2/PML transactivation (F003, F005); RUNX1 coactivation separable from HAT (P 19264921)
Dominant-negative interference	The seed's distinguishing sub-claim — truncated protein poisons WT complexes	Structurally plausible (retained PHD/HAT could occupy chromatin/complexes) but unproven
Gain-of-function neomorph	The seed's alternative sub-claim	Speculative; no supporting functional data found
Missense/HAT-domain hypomorph mechanism	Parallel mechanism for non-truncating variants	KAT6A is also missense-constrained (mis_z 3.69); not addressed by the truncation branch

The most parsimonious reading is that **haploinsufficiency + C-terminal partial loss-of-function** explains the late-truncating severity gradient without invoking a dominant-negative or neomorph. The seed hypothesis's added value is precisely the DN/GoF claim — which is exactly the part that lacks evidence.

Discriminating Tests

- 1. Patient-RNA NMD assay (highest priority).** Stratify patient fibroblasts/iPSC-neurons by variant class (early exon 1–15 vs late exon 16–17). Allele-specific RNA-seq ± cycloheximide.

Expected if hypothesis true: late-truncating transcripts retained near WT levels; early-truncating transcripts depleted.

2. **Mutant protein detection.** Targeted mass-spec / C-terminal-aware Western on the same cells. *Expected:* stable truncated protein only in the late-truncating group.
3. **Trans-interference reporter assay.** Co-express WT + truncated KAT6A on RUNX1/TBX1-dependent reporters and endogenous target ChIP. *Expected if dominant-negative:* truncated protein suppresses WT activity below the 50% dosage baseline. *If simple partial-loss:* activity tracks WT dosage, no extra suppression.
4. **Isoallelic mouse/organoid comparison.** Knock-in an exon-17 truncation vs a null allele; compare severity. *Expected if branch is real:* truncation allele more severe than null at matched dosage.
5. **Harmonized cohort meta-analysis** with uniform severity scoring and variant re-annotation to produce a pooled effect size and test for a distal-to-proximal severity trend within exon 16.

Curation Leads (require curator verification)

Candidate status/framing: Retain **ALTERNATIVE**, but annotate that the *clinical severity gradient* sub-claim is essentially **established** (three-cohort replication) while the *dominant-negative/gain-of-function mode of action* sub-claim is **speculative/unconfirmed**. Consider splitting the hypothesis into (a) a supported severity-modifier edge and (b) an unresolved mechanism-of-action edge.

Candidate **evidence** **references** + **snippets** **to** **verify:** -

P 30245513

— "late-truncating pathogenic variants (exons 16-17) are significantly more prevalent" -

P 35892268

— "Truncating variants in the last two exons of KAT6A were associated with poorer communication, daily-living skills, and socialization outcomes." -

P 38741077

— "late-truncating variants associated with a more severe form of intellectual disability" -

P 19264921

— "MOZ HAT activity is not required either for its role as Runx1 coactivator" -

P 22921202

— "lack of the histone acetyltransferase MOZ (MYST3/KAT6A) phenocopies DiGeorge syndrome" -

P 25912687

— "Mesp1-cre-mediated deletion of Moz results in high penetrance of VSDs"

Candidate pathophysiology nodes/edges: - Node: "NMD-escaping late-truncated KAT6A transcript (exon 16–17 PTC)" → Edge: "retains MYST-HAT catalytic domain; deletes C-terminal RUNX1-2 (aa 1913–1948) + PML/RUNX1 (aa 1517–1741) transactivation." - Edge (baseline): "KAT6A LoF (pLI 1.0, LOEUF 0.079) → haploinsufficiency → core ARTHS features." - Edge (unconfirmed): "late-truncated protein → dominant-negative interference with WT KAT6A/RUNX1 complexes" — flag as **needs direct perturbation evidence**.

Candidate ontology terms: histone H3 acetylation (GO:0043966); MYST-type HAT activity; RUNX1/CBF β transcriptional coactivation; nonsense-mediated decay (GO:0000184); neurodevelopmental/intellectual-disability phenotype terms (HPO).

Candidate knowledge_gaps / discussion prompts: - "No patient-tissue demonstration of truncated KAT6A protein or NMD escape as of 2026-09-07." - "Dominant-negative vs C-terminal partial-loss remains undistinguished." - "GenCC/ClinGen dosage-mechanism curation for the late-truncating branch: unverified (not queried this run)."

Provider artifact bundle: Computational lookups (Ensembl transcript coordinates, UniProt domain map, gnomAD constraint) and the PubMed search logs supporting F001–F005 should be deposited under `kb/hypotheses/Arboleda-Tham_Syndrome/late_truncating_nmd_escape/openscientist_artifacts` with a `MANIFEST.yaml` recording the retrieval date (2026-09-07), the exact accessions (ENST00000265713, Q92794, ENSG00000083168), and the rule-based (not assayed) status of the NMD prediction. No patient-level or controlled data were accessed; none should be bundled.

Evidence Base (literature summary)

PMID	Title (abbreviated)	Role in this report
30245513	KAT6A Syndrome genotype–phenotype, n=76	Primary support, severity gradient
35892268	Speech/language genotype–phenotype, n=49	Independent replication
38741077	Neuropsychological profile, n=15	Third-cohort replication
19264921	MOZ HAT/RUNX1 coactivation domains	Mechanism route (separable coactivation)
22921202	MOZ–Tbx1 / DiGeorge phenocopy	Competing haploinsufficiency
25912687	Moz mesodermal deletion → VSDs	Competing haploinsufficiency
25605372	Moz haploinsufficiency, B-cell/Meis1	Competing haploinsufficiency
41702672	Kat6a+/- mouse, acetyl-carnitine rescue	Competing model + therapeutic lead
12771199	RUNX1/MOZ MIP-1α promoter	C-terminal coactivation context
27939639	BRPF1 frameshift → truncated protein	Analogy for NMD escape in a partner gene
18754862	MOZ in normal/malignant hematopoiesis	Review-level orientation
18224409	AML1 complex chromatin regulation	Review-level orientation (PML/MOZ context)
16917507	HIPK1/2 in AML1/p300 transcription	Review-level orientation
33552646	Novel KAT6A frameshift, pancraniosynostosis	Phenotype breadth
29806701	Neonatal leukaemia (KAT6A-CREBBP)	Somatic KAT6A context (orientation)

Proposed Follow-up Experiments / Actions

1. **Run the patient-RNA NMD assay and mutant-protein detection** (Discriminating Tests 1–2) — closes the keystone gap and would convert the hypothesis from "partially supported" to "supported" or "refuted."
 2. **Perform the trans-interference reporter assay** (Test 3) to distinguish dominant-negative from simple C-terminal partial-loss — the single most decisive experiment for the seed's distinguishing claim.
 3. **Commission a harmonized three-cohort meta-analysis** (Test 5) to produce a pooled, quantitative severity effect size and test for a within-exon-16 proximal-to-distal gradient.
 4. **Query GenCC and ClinGen** with recorded version/date to convert Gap 5 from "unverified" to a documented positive/negative curation finding.
 5. **Deposit the computational artifact bundle** (Ensembl/UniProt/gnomAD lookups + PubMed logs) with a checksummed `MANIFEST.yaml` and explicitly flag the NMD prediction as rule-based, not assayed.
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*Report prepared for the Disorder Mechanisms Knowledge Base. Verdict: the Late-Truncating NMD-Escape Branch is **PARTIALLY SUPPORTED** — a reproducible clinical severity modifier with a structurally sound NMD-escape premise, but an experimentally unproven dominant-negative/gain-of-function mode of action layered atop an established haploinsufficiency baseline.*
